

Chapter 5

Locative and existential predication contrasts in Gawarbatī (Indo-Aryan) and the surrounding region

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This paper analyses the morphosyntactic variation in locative (LOC) and locational-existential (LOC-EX) clauses in Gawarbatī, an underdescribed Indo-Aryan language which has no dedicated formal marking of the LOC vs. LOC-EX contrast. The analysis also compares figure-ground predications in geographically adjacent languages. The results show that there are three morphosyntactic parameters which reflect, to varying degrees, the LOC vs. LOC-EX status of the predication: word order, indefiniteness marking and the lexical identity of the predicate. The parameter reflecting the LOC vs. LOC-EX alternation most consistently is word order. However, as shown in the corpus study of Gawarbatī, what is primarily encoded by word order variation is a range of available information-structural patterns, and they do not always easily match with the LOC vs. LOC-EX distinction.

1 Introduction

To describe the physical location of one object (*figure*) with respect to another object (*ground*), languages commonly use two types of clauses (see Clark 1978, Freeze 1992 and Hengeveld 1992, among others):

- locative predications (LOC), e.g. *the book is on the table*,
- locational-existential predications (LOC-EX), e.g. *there is a book on the table*.



When referring to both LOC and LOC-EX types of clauses, we use the term *figure-ground predications*.¹

The first aim of this paper is to contribute to the research on figure-ground predications by focusing on the problem of distinguishing between LOC and LOC-EX clauses in languages where the LOC vs. LOC-EX contrast is not clearly reflected in their morphosyntax. There is a large number of such languages; for instance, Creissels (2019b) shows that more than half of the world's languages lack a dedicated LOC-EX construction such as, for example, *there is* in English. Although the LOC vs. LOC-EX distinction is to some extent reflected in the morphosyntactic variation found in figure-ground predications (in particular, definiteness marking and word order), we will mostly address cases where there is no dedicated marking differentiating LOC and LOC-EX clauses.

The second aim of this paper is to bring more data from non-European low-resource languages to the discussion on figure-ground predications. We mainly focus on Gawarbati, an understudied Indo-Aryan language, but also discuss the languages of the surrounding area, namely the Hindu Kush-Karakoram region (henceforth, Hindu Kush). The comparison of Gawarbati with its geographical neighbours also allows us to contribute to the discussion on the diversity and development of LOC and LOC-EX clauses in an areal-typological perspective, cf. Zeitoun et al. (1999) on Formosan languages, Bentley et al. (2015) on Italo-Romance varieties and Creissels (2019a) on the languages of the Sudanic belt. The data for the study of figure-ground predications in Gawarbati come from a corpus of spoken language and are complemented by some directly elicited material. The data for the study of figure-ground predications in other Hindu Kush languages are mainly collected by using existing descriptive literature.

The research questions of the paper can be formulated more precisely as follows:

- What types of figure-ground predications can be distinguished in Gawarbati, based on formal grounds?
- How do these types correspond to the semantic/pragmatic criteria for differentiating LOC and LOC-EX clauses cross-linguistically, as discussed in the literature? (These criteria are described in §2.)
- What areal patterns in figure-ground predications are found in the Hindu Kush region, and to what extent can the results of the Gawarbati study be generalised to other languages of the area? Can any of the observed patterns be explained in terms of language relatedness or language contact?

¹Haspelmath (2025) groups these two types of predications under the general term *locational clause*.

The structure of the paper is the following. §2 provides a theoretical background on the difference between LOC and LOC-EX clauses. §3 introduces the Gawarbat language and other languages of the Hindu Kush. §4 describes the study of figure-ground predications in Gawarbat. §5 is dedicated to an areal survey. In §6, the main findings of the paper are summarised and discussed from a theoretical and typological perspective.

2 Theoretical remarks

In the literature, the precise difference between LOC and LOC-EX predications remains a matter of debate. Table 1 summarises three common approaches to defining the LOC vs. LOC-EX contrast.

Table 1: Approaches to differentiating LOC and LOC-EX predications

Parameter	LOC	LOC-EX
Information structure	figure = topic	figure = focus
Definiteness	figure is definite	figure is indefinite
Perspectivisation	the Perspectival Center is the figure	the Perspectival Center is the ground

According to Koch (2012), the main criterion for differentiating between LOC and LOC-EX clauses is information structure. In LOC clauses, the figure is the topic of the sentence, while in LOC-EX clauses, the figure is the focus of the sentence.

An alternative approach, recently proposed by Haspelmath (2025), suggests that the criterion for distinguishing LOC and LOC-EX predications is definiteness of the figure: when the figure is definite, it is a LOC predication, and when the figure is indefinite, it is a LOC-EX predication.

The third influential view found in the literature suggests that the difference between LOC and LOC-EX predications is determined by perspectivisation, a parameter distinct from information structure or definiteness and also underlying, for example, the choice between active and passive voice (Borschev & Partee 2002, Partee & Borschev 2007, Creissels 2019b: 49–50). The key element of the perspective structure is the Perspectival Center, i.e. “the participant chosen as the point of departure for structuring the situation” (Borschev & Partee 2002: 124): in LOC the Perspectival Center is the figure, in LOC-EX the Perspectival Center is the ground.

Perspectivisation is a rather abstract notion that has to do with our perception and categorisation of the world, but, to some extent, it can be operationalised through the notion of presupposition. As noted by Borschev & Partee (2002), an important property of the Perspectival Center is that it must be presupposed to exist, while the other participant does not have to be presupposed. The obligatory presupposition of existence for the Perspectival Center has implications for the formation of negative LOC and LOC-EX clauses. Since the presupposition cannot be cancelled under negation, in LOC clauses it is not possible to negate the existence of the figure (the Perspectival Center): the scope of negation includes only the ground, cf. *the book is not on the table*. By contrast, in LOC-EX clauses the existence of the figure can be cancelled under negation, cf. *there are no books on the table*.

Obviously, the three parameters presented in Table 1 correlate with each other. For example, indefinite non-specific referents typically lack the presupposition of existence; topical nominals are usually definite. However, we also find examples of discrepancies between the parameters (Partee & Borschev 2007, Creissels 2019b: 47–50, Haspelmath 2025). For instance, the figure in (1) is definite but focal, the figure in (2) is indefinite but topical.

- (1) THE DOG is on the bed. (Haspelmath 2025)
- (2) Russian (Indo-European, [russ1263]; based on Paducheva 2008: 150)²
Kreslo / na verande est' \.
 chair on porch.LOC.SG be.PRS
 (If you are looking for chairs) ‘There is a chair on the porch.’

In both examples, the ground seems to be associated with a greater presuppositionality of existence than the figure, although the Perspectival Center in affirmative clauses is not always perfectly clear and its annotation usually requires more context (see also discussion in §4.2.1).

This paper mainly follows the core vs. periphery approach, i.e. clauses which can be defined as LOC and LOC-EX according to all three parameters are considered core LOC and LOC-EX clauses, whereas all other clauses are considered peripheral. In the corpus study (§4.2), we do not immediately merge the three parameters into one single category. Instead, we consider them separately, so that the results of the study can be interpreted in the light of each one of the described approaches.

In the rest of the paper, we focus on the morphosyntactic variation in figure-ground predications and its relation to the LOC and LOC-EX distinction, using the data from Gawarbati and the languages of the surrounding region.

²Slashes in this example indicate intonation contour.

3 Gawarbati and other languages of the Hindu Kush

3.1 Gawarbati and Gawarbati grammar

Gawarbati [gawa1247] is an Indo-Aryan language spoken by approximately 20,000 people in a number of villages scattered along the Kunar River in the mountainous border area of Pakistan and Afghanistan (Fazal Akbar, p.c.), in Pakistan's Lower Chitral District, Khyber Pakhtunkhwa Province, and in Afghanistan's Nari District, Kunar Province. The approximate location of the Gawarbati-speaking area can be seen in Figure 2 at the beginning of §5.1. According to the genealogical classification suggested in Bashir (2003), Gawarbati together with its two near-extinct varieties Shumashti [shum1235] and Grangali (in Afghanistan, [gran1245]) and the neighbouring language Dameli (in Pakistan, [dame1241]) constitute a Kunar group of Indo-Aryan, although the exact classification of Dameli remains uncertain (Strand 2001: 258, Halfmann 2022: 9–10).

The available literature on Gawarbati grammar is limited to three papers (Grierson 1919, Morgenstierne 1950, Kohistani 1993), based on first-hand data, and two short descriptions by Edelman (1983) and Bashir (2003). The mentioned works together with the currently ongoing but not yet published corpus study of Gawarbati provide insight into the main morphosyntactic characteristics of the language, some of which are briefly discussed below.

Plural marking in Gawarbati is optional and occurs rather rarely in spontaneous speech. The case system includes (unmarked) Nominative, Oblique, Genitive/Ablative, Dative and Ergative/Instrumental cases. The Oblique case encodes direct objects, cf. *hetsi-a* [cow-OBL] in (3), as well as locative and temporal arguments. The Oblique is also the most frequent case used for nouns in postpositional phrases, cf. *mu^ha-a* [tree-OBL] in (3). The numeral *jak/ja* 'one' (the distribution between the two variants is not clear), also illustrated in (3), can be used as an indefinite article (see a more detailed discussion in §4.2.2).

- (3) *hetsi-a ja mu^ha-a manzi gent-us*
 cow-OBL INDF tree-OBL with/at tie-PST.3SG
 '[He] tied the cow to a tree.'³

In terms of alignment typology, Gawarbati case marking displays a tripartite system, as seen in the distinct case marking of S, A and P in (4)–(5).

³If not stated otherwise, all Gawarbati examples come from the corpus.

- (4) *mon-a se muma: dzo pe-san bua*
 my-M DIST.SG[NOM] uncle[NOM] PTCL sleep-PTCP be.PST.3M.SG
 ‘My uncle was sleeping.’
- (5) *tuī asa rupai-a kenna a:n-u*
 you.SG.ERG PROX.OBL.SG rupee-OBL where.from bring-PST.2SG
 ‘Where did you bring this money from?’

However, the TAM-based ergativity split and differential object marking often lead to neutralisation of these distinctions. The Ergative marking of A is limited to perfective TAM categories, whereas in imperfective TAM categories, A appears in the Nominative form. P is marked by the Oblique case only when it is definite; otherwise it remains in the Nominative form. In verbal agreement, Gawarbati follows a Nominative-Accusative pattern, thus showing different alignment types in head marking and dependent marking.

The TAM system of Gawarbati is composed of both synthetic and periphrastic forms and includes some modal constructions. Non-verbal predications are formed with a copula which cannot be left out in any of the tenses (see also Bashir 2023: 262). The arguments of predicates, by contrast, are freely omitted if they can be inferred from the previous context.

The unmarked basic word order in Gawarbati is SOV, but it is flexible and heavily dependent on information structure. The topical constituents appear in the clause-initial position, and focus is strongly associated with the immediately preverbal position. For example, in (6) the topic is *sauda:* ‘goods’. It is moved to the beginning of the sentence, while the preverbal position is occupied by the focal argument *lawans-ana* ‘from Drosh’.

- (6) [*sauda:*]_{topic} *ama [lawans-ana]_{focus} a:n-imek*
 goods.NOM/OBL we Drosh-GEN/ABL bring-PRS.1PL
 (From where do you bring the stuff for your shop?) ‘We bring the stuff from Drosh.’

Gawarbati employs a few particles such as *dzo* in (4) above, but none of them is uncontroversially associated with any information structure category.

3.2 Gawarbati documentation and corpus building

The data used in this study come from a spoken corpus which is currently being compiled. It is part of a larger, ongoing project, documenting Gawarbati in close collaboration with the speaker community. The project was launched in 2021 and

consists at the time of writing of a Swedish-Pakistani 6-member team:⁴ Henrik Liljegren (principal investigator and project leader), Anastasia Panova (research assistant/doctoral student), Fazal Hadi (local team coordinator), Allauddin Torwali (deputy local team coordinator), Fazal Akbar (language consultant, mainly responsible for lexicographic data collection and processing), Nasirullah Nasir (language consultant, mainly responsible for recording and processing meta data) and Abdullah Soan (language consultant, mainly responsible for annotation of recorded data). The activities in Pakistan take place under the auspices of the Forum for Language Initiatives (FLI), a regional language resource centre founded in 2002, based in Islamabad (Liljegren & Akhunzada 2017: 72–77). The aim of this project is to collect, annotate and analyse language data; to produce a typologically informed grammatical description; to compile a lasting record of the language in the form of an annotated audio and video corpus (eventually based on at least 20 hours of recordings); and to build a lexical database.

The version of the corpus used in the study of figure-ground predications comprised 40 texts (=recorded events) altogether, recorded in 2021–2022, in the village Arandu in Pakistan, then segmented, transcribed and given an utterance-by-utterance free translation, first into Urdu (Indo-European, [urdu1245]), and then into English, by the team members affiliated with FLI. Subsequently, it was morphologically analysed and glossed by Anastasia Panova, based on the former annotation.⁵ It consists of about 38,000 tokens, and corresponds to four hours of recordings.

3.3 The genealogical and areal context of Gawarbati

Gawarbati and the other languages of the Kunar group (mentioned above) are included into a cluster of Indo-Aryan, often referred to as Dardic. However, it should be noted that “Dardic” or “Dardic languages”, since Morgenstierne’s groundbreaking work (1961: 138–139), should be understood as geographic or areal labels rather than genealogical terms. It is applied to the many Indo-Aryan languages spoken in the Hindu Kush region, a mountainous territory comprising northeastern Afghanistan, northernmost Pakistan and the northern parts of Kashmir and Ladakh. To stress the areal affinities between these languages rather than enforcing a misguided idea of a genealogical designation or intermediate node on the Indo-Aryan tree, Liljegren (2016: 13–16) has suggested the

⁴Note that there are 7 people involved, whereby the team coordinator position was filled by one person initially then taken over by another.

⁵The translations presented in the paper rely more on glosses than on translations extracted from the corpus, as the latter were often not precise enough.

term “Hindu Kush Indo-Aryan (HKIA)” to replace Dardic. At least 25 distinctly different Indo-Aryan languages have been at home in this particular region for an extended period of time. Those can in their turn, following Bashir (2003: 824–825) or Strand (2001: 258), be grouped into smaller relatedness clusters or groups, such as Pashai, Kunar/Pech, Chitral, Kohistani, Shina and Kashmiri. In addition to languages with a longstanding presence in the region, we also find a number of Indo-Aryan varieties that have closer genealogical and historical ties with languages of the Indo-Pakistani plains (Liljegren 2017: 110).

Apart from Indo-Aryan, which is the numerically dominant group of the Hindu Kush, five more phylogenetic groups are represented in this region: Iranian, Nuristani, Sino-Tibetan, Turkic and the language isolate Burushaski [buru1296]. As shown by Liljegren (2020: 217–221), the languages of the Hindu Kush have phonological and lexico-grammatical features that are shared by languages forming a core of the Hindu Kush, regardless of phylogenetic identity, such as large sets of retroflex consonants, utterance-final question markers and vigesimal numeral bases. In addition to that, there is a significant level of subareality apparent throughout the Hindu Kush, with more local feature clustering in geographical subregions (Liljegren 2020: 221–225). The latter appears to be particularly prominent in the western part of the Hindu Kush, related to the fact that the ethnolinguistic communities in Nuristan, situated in the remote mountainous parts of northeastern Afghanistan and the immediately surrounding areas, did not become part of the larger Islamic world until very recently. Until then, these communities formed a relatively tight-knit cultural unit vis-à-vis the already Islamised outside world, linguistically reflected in e.g. the presence of retroflex/rhotacised vowels, a certain type of kin-term polysemy pattern, pronominal kinship suffixes, geomorphicity, and bisyndetic contrast marking (Di Carlo 2011, Heegård & Mørch 2004, Heegård & Liljegren 2018, Liljegren 2021, Liljegren & Svärd 2017). Gawarbatī finds itself at the outskirts of this subarea, thus sharing some, but not all, of those characteristics.

All of this indicates that language contact has played an important role, both in the larger region, and in the immediate surroundings of Gawarbatī. Hence, studying and paying attention to the areal-typological context is essential to understand the development of particular linguistic phenomena in the Hindu Kush languages, a matter we will have reason to return to in further detail in §5.

4 Figure-ground predications in Gawarbatl

4.1 The basic morphosyntax of LOC or LOC-EX predications

Direct elicitation of LOC or LOC-EX predications from English (or Urdu) to Gawarbatl gives quite straightforward results. Both types of clauses involve a nominal phrase, a locative phrase and the copula. The difference between predications lies primarily in word order.⁶ We can see that in the LOC clause (7a), the figure *tekura* ‘boy’ precedes the ground *ba:y* ‘garden’, whereas in the LOC-EX clause (7b), the figure follows the ground. In addition, the figure in the LOC-EX clause (7b) is modified by the marker *jak* which seems functionally similar to the indefinite article in the English stimulus.

- (7) a. *tekura ba:y-a t^hana*
 boy garden-OBL be.PRS.3M.SG
 ‘The boy is in the garden.’ [elicited]
 b. *ba:y-a jak tekura t^hana*
 garden-OBL INDF boy be.PRS.3M.SG
 ‘There is a boy in the garden.’ [elicited]

Despite the different positions in the clause, the figure seems to be the subject in both types of predications, as it appears in the Nominative case and controls the copula agreement, cf. (7) and (8) where the copula agrees with the figure in the masculine gender and in the feminine gender, respectively.

- (8) a. *uzal-i dzama:it la:m-an-a manzi t^hini*
 white-F mosque village-GEN-M inside be.PRS.3F.SG
 ‘The white mosque is in the middle of the village.’ [elicited]
 b. *la:m-an-a manzi jak dal dzama:it t^hini*
 village-GEN-M inside INDF big mosque be.PRS.3F.SG
 ‘There is a big mosque in the middle of the village.’ [elicited]

Negative LOC-EX clauses are formed with the default negative particle *na* (9). In addition, we find one more possible strategy of deriving negative LOC-EX clauses in the corpus, namely with the negative predicative particle *nahn-* which shows agreement with the Figure 10.

⁶Word order should here, and henceforth, be understood as the order of syntactic constituents.

- (9) *ba:y-a kara xalaq na t^hanet*
 garden-OBL any people NEG be.PRS.3PL
 ‘There are no people in the garden.’ [elicited]
- (10) *ite wa:ri xalaq nahn-et*
 here other people NEG.be-3PL
 ‘There are no other people here.’

All of the examples provided so far included the Present tense form of the copula. Table 2 shows the full paradigm of the copula in the Present as well as in the Past tense (copular forms in other tenses are also attested in the corpus but they rarely occur in LOC and LOC-EX clauses). In the Past tense, the stem is *b-*, see also (11). The original meaning of the verb stem *b-* is ‘become’, but synchronically the forms of *t^han-/t^hin-* in the Present and the forms of *b-* in the Past constitute subparts of one and the same paradigm.⁷

Table 2: The Gawarbati copula in the Present and Past tenses

	Present			Past		
	SG.M	SG.F	PL	SG.M	SG.F	PL
1	<i>t^hanem</i>	<i>t^hinim</i>	<i>t^hanek</i>	<i>bom</i>	<i>bum</i>	<i>bok</i>
2	<i>t^hanes</i>	<i>t^hinis</i>	<i>t^haneu</i>	<i>bos</i>	<i>bus</i>	<i>bo</i>
3	<i>t^hana</i>	<i>t^hini</i>	<i>t^hanet</i>	<i>bua</i>	<i>bui</i>	<i>bot</i>

- (11) *madina-an-i rija:sat-a xo insa:f bui*
 Madinah-GEN-F state-OBL PTCL justice be.PST.3F.SG
 ‘There was justice in the state of Madinah.’

The copula is also used in possessive predications, which are structurally similar to LOC-EX with the exception of case encoding — the possessor is usually encoded by the Genitive case (12).

- (12) *tasan-i sat zu bot*
 DIST.GEN.SG-F seven daughter be.PST.3PL
 ‘He had seven daughters.’

⁷The stem *t^han-/t^hin-* also has a shortened variant *t^h-* to which regular Present tense affixes attach, cf. (21) below. Those compete paradigmatically with the forms of *t^han-/t^hin-*, although they are much less frequent.

Although copula-based clauses represent the primary strategy for describing figure-ground relations, there is arguably an alternative strategy available for LOC predications that does not involve the copula. In the corpus, we find recurrent cases of the verb *niç-* ‘sit (down)’ in LOC predications: sometimes it is intended as a posture verb but often used in a more abstract sense ‘be (located in)’. For example, in (13), the interviewer says that the interviewee and the interviewer himself are “sitting” at the upper mill. However, it is clear from the video recording that at the moment of saying that, both participants are in fact standing in front of the entrance to the watermill. This contradiction shows that the verb *niç-* can indicate physical presence without specifying posture.

- (13) *antar dʒant-a niç-isən tʰanek*
 upper water.mill-OBL sit-PTCP be.PRS.1PL
 ‘We are at the upper mill.’ (lit. [We] are sitting at the upper mill.)

It is noteworthy, however, that when speakers were asked to translate examples like (13) into English without context, they consistently favoured the literal translation with ‘sit (down)’ over the more abstract translation with ‘be (located in)’.

Another semantic extension of *niç-* ‘sit (down)’, less relevant for the present paper but confirming the gradual semantic bleaching of the verb, is illustrated in (14), where it is used in the sense of ‘live’.

- (14) *time tine niç-isən bot*
 DIST.PL there sit-PTCP be.PST.3PL
 (Allah gave them a good place) ‘They started living there.’

Having outlined how figure-ground predications are structured, we will now provide a usage-based account of these types of clauses.

4.2 A corpus study: towards a bottom-up typology of figure-ground predications

4.2.1 Annotation

The aim of the corpus study was to identify the types of figure-ground predications in Gawarbatl using natural discourse data and then examine how these types relate to the LOC vs. LOC-EX contrast.

After a manual check of all the sentences in the corpus, the clauses describing temporary or permanent relations between figure and ground were selected for

further analysis. We did not include clauses involving the negative predicative particle *nahn-* as dedicated negative existential constructions often behave very differently across languages (Veselinova 2013). We also excluded clauses with predicates other than the copula, i.e. with the verb *nic-* ‘sit (down)’ (see §4.1), clauses with omitted subject, as in (15), and clauses describing abstract spatial relations and time as space metaphors, as in (16).

- (15) *nori te muɬ^ha-a manzi t^hanem*
 now DEM.OBL.SG tree-OBL inside be.PRS.1M.SG
 ‘[I] am now inside this tree.’
- (16) *amn-a te la:m-a ja cil-i wax bui ze*
 our-M DEM.OBL.SG village-OBL INDF such-F time be.PST.3F.SG COMP
 ‘There was a time in our village when...’

The final sample included 87 examples, of which 74 were affirmative clauses and 13 were negative clauses.

All the examples included in the sample were annotated for a number of parameters.⁸ First, we analysed formal variation across the clauses, specifically variation in word order and the presence of the marker *ja/jak* – these were the main features which varied across the examples on the structural level. Second, we also annotated the sample according to the parameters which have been discussed as relevant with respect to the LOC vs. LOC-EX contrast in the previous literature, namely definiteness of the figure, information structure, and perspectivisation.

In the annotation of the definiteness of the figure, three categories were distinguished – definite, indefinite specific and indefinite non-specific, the latter being associated primarily with negative contexts like *there are no books on the table*.

For encoding information structure, we used topic vs. comment and focus vs. background distinctions, following Krifka (2008) and Aissen (2023) as the main references.⁹ Both figure and ground were encoded as being (a part of) one or several information structure categories. For example, in (17), the figure is the topic, and the ground is part of the comment, while in (18), the figure is both the focus and part of the comment, and the ground is both the topic and part of the background.

⁸The annotated dataset is available at https://osf.io/bq98g/?view_only=a61f393ae43649f48a8501a8fe2fd735.

⁹In the annotation, we also differentiated between plain and contrastive versions of topic and focus, but for the purposes of this paper these distinctions were merged.

- (17) [ama]_{topic} [dam haranu-a t^hanek]_{comment}
 we Dam Arandu-OBL be.PRS.1PL
 ‘We are in Arandu Dam.’
- (18) [ate hadika-a]_{topic} [sik-a:n bot]_{comment}
 [ate hadika-a]_{background} [sik-a:n]_{focus} [bot]_{background}
 here place-OBL Sikh-PL be.PST.3PL
 (Actually, it was not Godibar.) ‘Sikhs used to live here.’

Some sentences, especially at the beginning of the recorded events, were annotated as *thetic* (comment-only), because none of their parts were introduced, either explicitly or implicitly, in the previous context.

Finally, to identify the *Perspectival Center*, we analysed the presupposition vs. assertion structure of the sentences. As mentioned in §2, the *Perspectival Center* is normally presupposed to exist, so if the figure lacks a presupposition of existence, the *Perspectival Center* is usually the ground. The clearest cases are negative clauses like (19), where the existence of the figure is negated. Hence, the *Perspectival Center* is the ground.

- (19) [ate dameɽ-a]_{PerspectivalCenter} iskul na t^hana
 here Damil-OBL school NEG be.PRS.3M.SG
 ‘There is no school in Damil.’

In affirmative clauses, the figure was encoded as not having a presupposition of existence when the previous context was insufficient to assume that it existed. For example, in (20), although the general subject of the utterance fragment — the earthquake — is clear from the previous context, the existence of the cave cannot be assumed. Therefore, the cave cannot be a *Perspectival Center*.

- (20) [ite aman-a mongoldam-an-a rataj ite hadika-a]_{PerspectivalCenter}
 here our-M Mongoldam-GEN-M on here place-OBL
 bi ɖal ya:r t^hana
 ADD big cave be.PRS.3M.SG
 (The places of worship were destroyed by the volcano. There are still large black rocks lying here. There is also an earthquake mountain here.)
 ‘There is also a large cave here in Mongoldam.’ (There is a big rock, and hot air comes from inside.)

The opposite situation, i.e. where the figure is annotated as a *Perspectival Center*, is illustrated in (21). In such examples both the figure and the ground can be

presupposed to exist, or the ground can be associated with a lesser presuppositionality of existence. This can be compared with the following example from Partee and Borschev (2007: 157), in which the existence of the ground is cancelled: *The students were not at the concert. There was no concert.*

- (21) [time]_{PerspectivalCenter} *te* *la:m-a* *t^h-imet* *dzo*
 DIST.PL DEM.OBL.SG village-OBL be-PRS.3PL PTCL
 (These people come to Arandu. When they come to Arandu,) ‘they are in this village because..’

It should be noted, however, that the perspective structure in affirmative sentences, generally, was the most difficult to annotate. The results of the annotation of the Perspectival Center should therefore be taken with caution.

Summarising the results of annotating definiteness, information structure and perspectivisation, we conclude that the groups of the examples identified by different parameters largely overlap. The diagram presented in Figure 1¹⁰ shows the proportion of the examples which can be identified as LOC and LOC-EX based on three parameters together (the red and green parts respectively), and what proportion of the examples has to be left out as instances of the periphery of the domain (the blue part).

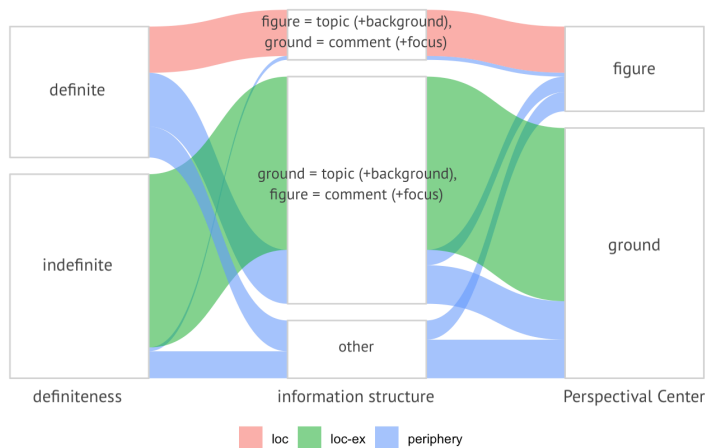


Figure 1: Combinations of annotated features in the sample and their respective proportions

¹⁰The diagram was created with R (Team 2022) package *ggforce* (Pedersen 2024).

Although the number of examples that ended up in the periphery might seem too big to represent the periphery (they constitute approximately one third of the sample), Figure 1 nevertheless shows that it is possible to identify core LOC and LOC-EX types of figure-ground predications on purely semantic/pragmatic grounds. Examples of clauses which can be considered LOC and LOC-EX according to all three parameters are provided in (22) and (23), respectively.

- (22) *se baloṭista:n-a bua*
 DIST.SG Balochistan-OBL be.PST.3M.SG
 ‘He was in Balochistan.’

- (23) *tasa manzi hast-a gila bua*
 DIST.OBL.SG inside hand-OBL thing be.PST.3M.SG
 ‘There was something in his hand.’

(24)–(25) are examples of discrepancies between the three parameters, and thus they are annotated as peripheral. In (24), the figures are definite but focal; in (25), the figure in the second clause is topical but lacks a presupposition of existence (see also (35) for the same type of discrepancy).

- (24) *haranu-a asan-a a:ma t^hana dokala:m-a*
 Arandu-OBL PROX.GEN.SG-M house be.PRS.3M.SG Dokalam-OBL
asan-i ta:nu sasi t^hini
 PROX.GEN.SG-F self’s sister be.PRS.3F.SG
 ‘In Arandu was his house, (whereas) in Dokalam was his sister’s (house).’
- (25) *te ki ki çai t^hanet time tine na t^hanet*
 here what what thing be.PRS.3PL DIST.PL there NEG be.PRS.3PL
 ‘What things exist here (in Pakistan) but do not exist there (in Afghanistan)?’ (lit. ‘What things are here that are not there?’)

The core vs. periphery approach confirms that LOC and LOC-EX categories as defined through definiteness, information structure and perspective structure alike are in principle relevant to the Gawarbatl grammar and discourse. As a next step, we checked whether there are any correlations between the parameters relevant to the LOC vs. LOC-EX contrast and the formal variation present in the examples, i.e. variation in definiteness marking and word order.

4.2.2 Results: the marker *ja/jak*

The sample contains only eight examples featuring the marker *ja/jak* which modifies the figure. Table 3 shows how the presence of *ja/jak* corresponds to the actual (in)definiteness status of the figure.

Table 3: The presence of *ja/jak* and definiteness of the figure in the Gawarbati sample

	<i>ja/jak</i>	∅
the figure is definite	0	34
the figure is indefinite specific	8	22
the figure is indefinite non-specific	0	23

In all examples containing the marker *ja/jak*, the figure is indefinite specific, cf. (26).

- (26) *mo manzi ite ja ba:ja t^hana*
 LOBL with here INDF brother be.PRS.3M.SG
 ‘There is a brother with me here.’ (His name is Hisham.)

It seems that the marker *ja/jak*, apparently grammaticalised from the numeral *jak* ‘one’, is indeed used only in indefinite specific contexts, thus not covering the whole indefinite domain (cf. the similar analysis of the article *áa(k)* in related Palula by Becker (2021: 132, 266–267)). Moreover, *ja/jak* often does not mark the figure even when it is indefinite specific, due to its incompatibility with plural referents and mass nouns. Finally, we find examples where *ja/jak* is absent, although there are no apparent reasons for it to be omitted, as in (27).

- (27) *ma:lai sudi bui*
 below ladder be.PST.3F.SG
 ‘There was a ladder below.’

These observations indicate that the marker *ja/jak* is not a fully grammaticalised indefinite article. Although its presence implies the indefinite status of the figure and, subsequently, the LOC-EX status of the clause (if we use definiteness to distinguish between LOC and LOC-EX), its absence does not imply anything with respect to the LOC vs. LOC-EX contrast.

Interestingly, the eight examples where *ja/jak* is found are also quite homogeneous in terms of perspective and information structures. In all of these cases,

the figure is part of the comment, while the ground is usually the topic. Furthermore, the Perspectival Center in these examples is always the ground. Thus, technically the presence of *ja/jak* in our sample predicts the LOC-EX status of the predication, regardless of the criteria we choose to distinguish between LOC and LOC-EX types of figure-ground constructions.

4.2.3 Results: word order

Overall, in the clauses included in the sample, the order of the figure (F), the ground (G) and the verb (V) follows one of three patterns: in 25 of the examples, the word order is F–G–V; in 59, it is G–F–V; and three of the examples display the order F–V–G. Word order is often motivated by the mechanism of anchoring, i.e. the tendency to start the utterance with a reference point and then relate the rest of the information to it (in the literature, anchoring has been primarily discussed for possessive constructions, see, e.g., Koptjevskaja-Tamm 2002, but it can also be applied to the clause level). Within this view, we expect participants which are definite, topical and presupposed to serve as better anchors, and therefore would appear at the beginning of the clause.

Table 4 shows how the order of the figure, the ground and the verb corresponds to definiteness of the figure.

Table 4: Word order and definiteness of the figure in the Gawarbat sample

	F–G–V	G–F–V	F–V–G
the figure is definite	17	16	1
the figure is indefinite specific	4	26	0
the figure is indefinite non-specific	4	17	2

An indefinite figure tends to occupy the second position, but a definite figure occurs almost as often at the beginning of the clause as in second position.

The correspondence between word order and information structure is shown in Table 5.

As already mentioned in §4.1, the relationship between word order and information structure in Gawarbat manifests itself in the topic-first principle and the existence of a preverbal focus position. Both of these patterns are observed in figure-ground predications. The figure and the ground tend to occur at the beginning of the clause when they are topical, and in preverbal position when they

Table 5: Word order and information structure in the Gawarbatı sample

	F-G-V	G-F-V	F-V-G
figure ∈ topic (+background); ground ∈ comment (+focus)	13	0	0
ground ∈ topic (+background); figure ∈ comment (+focus)	6	51	2
other	6	8	1

are focal. This is illustrated in (28)–(29), provided with a partial annotation of information structure.

- (28) [grond]_{topic} [lau dura:je]_{focus} t^hini
 ground very far be.PRS.3F.SG
 (Why don't you play on the ground? Why do you play on the road?) 'The ground is far away.' (The road is near.)

- (29) ā: ki xabar ze [te]_{topic} [me]_{focus} t^haneu
 I what news COMP here you.PL be.PRS.2PL
 (I thought it was a closed area, then) 'I found out that you were here.'

In line with the principles described above, there are no examples with a topical figure in the preverbal position. At the same time, surprisingly, the sample includes six examples with a topical ground immediately preceding the verb. For instance, in (30), the ground *tine* 'there' refers to the place discussed in the previous context and functions as a plain aboutness topic, but it nevertheless follows the figure which, in turn, presents a new piece of information about the ground.

- (30) [lutr-a sum-na:m]_{comment} [tine]_{topic} [t^hanet]_{comment}
 red-M soil-PL there be.PRS.3PL
 (...whenever they dig, in that place they [long pipes] come out.) 'There is a lot of red soil there.' (They brought these pipes, and they used to make pots.)

There is a possibility that the intended information structure, in examples like (30), is expressed by intonation rather than by word order, but a systematic investigation of intonation in Gawarbatı is outside the scope of this study.

Regarding the examples showing “other” types of information structure, several of them were grouped separately because of the presence of a temporal argument serving as the topic of the clause, cf. (31). Consequently, both the figure and the ground in such examples are included in the comment.

- (31) *au laka [a:inda-a şatsi]_{topic} [tu ate t^hanes]_{comment}*
 and like future-OBL for you.SG here be.PRS.2SG

(A: We haven’t seen such good doctors who would come and work here anymore. After him, we have had only young people. B:) ‘And, like, in the future, you will be here.’

In several examples, the verb represents a separate information structural unit, cf. (32) where the verb is focused and contrasted with the negated verb in the next clause. In such examples, both the figure and the ground belong to the background part.

- (32) [*aman-i la:m-a primeri zana:na sikul*]_{background} [*t^hanet*]_{focus}
 our-F village-M primary woman school be.PRS.3PL
 ‘In our village, the primary girl schools exist,’ (while the middle high schools do not.)

In addition, a few examples from the “other” category arethetic sentences.

It is also shown in Table 5 that the ground occurs after the verb in three examples. On the one hand, the clause-final occurrences of the ground can be afterthoughts, i.e. fragments that do not necessarily share the syntactic and information structures with the rest of the clause. On the other hand, it is possible that Gawarbatl has a certain information structure (sub-)category associated with the post-verbal position. The latter analysis has previously been proposed for Urdu. According to Butt & King (1996), the post-verbal position in Urdu is reserved for not-new and non-prominent information. Likewise, in Gawarbatl the constituents that come after the verb normally convey information that can be reconstructed from the context. For instance, in (33), the fact that the speaker is talking about the Arandu valley is evident from the part of the discourse preceding the example (not provided here due to space restrictions).

- (33) *bidzli-an-a sahulat t^hana gal fa:r*
 electricity-GEN-M facilities be.PRS.3SG.M valley yonder
 ‘There is electricity in the valley.’

To sum up, the word order in figure-ground predications generally follows the information structure, although there are exceptions. This observation, in turn, suggests that if we choose information structure as the criterion for distinguishing LOC and LOC-EX clauses, word order can be considered a formal expression of the typology of figure-ground constructions. At the same time, since there are more than two possible information structures and more than two possible word order patterns available in figure-ground predications, the resulting typology must be significantly more complex than simply a LOC vs. LOC-EX dichotomy.

Finally, moving on to the correspondences between word order and perspective structure, we observe a tendency to start the clause with the Perspectival Center, cf. Table 6.

Table 6: Word order and perspective structure in the Gawarbatı sample

	F-G-V	G-F-V	F-V-G
Perspectival Center: figure	16	5	1
Perspectival Center: ground	9	54	2

For example, in (34), the ground *tine* ‘there’ has the presupposition of existence, while the figure does not, and therefore the clause starts with the ground.

- (34) [*tine*]_{PerspectivalCenter} *za:tak* *bot*
there young.people be.PST.3PL
(I saw the defense checkpoint near the bridge.) ‘There were boys there.’

A closer examination of the exceptions to this tendency reveals two patterns. First, the exceptions include the six sentences already discussed above as examples of discrepancies between word order and information structure — apparently, the word order in these sentences is influenced by some other factors, not included in the annotation. Second, in some of the examples where the Perspectival Center occupies a position other than the first place, the word order can be explained in terms of information structure. For instance, the figure in (35) lacks a presupposition of existence, and therefore it cannot be the Perspectival Center. Instead, it is the topic of the sentence, and that is the reason for its appearance at the beginning of the clause.

- (35) *ime* *cil-a* *telar* [*te*]_{PerspectivalCenter} *na* *t^hanet*
PROX.PL such-M tailor here NEG be.PRS.3PL
(How much does a good tailor charge? <...>) ‘There are no such tailors here.’

From that, we conclude that the word order in the examples of our sample can be accounted for by information structure much more readily than by perspective structure.

4.2.4 Summary

The corpus study has shown what the bottom-up typology of figure-ground predications in Gawarbat looks like. Based on formal grounds, i.e. the presence of the marker *ja/jak* and word order, six groups of examples can be distinguished (although one of the groups lacks examples altogether), see Table 7.

Table 7: The formal typology of figure-ground predications in the Gawarbat sample

	<i>ja/jak</i>	$\emptyset$
F–G–V	1	24
G–F–V	7	52
F–V–G	0	3

The variation shown in Table 7 is related to the LOC vs. LOC-EX contrast as defined through different parameters in a rather complex way. The presence of *ja/jak* predicts the indefinite status of the figure (although far from all indefinite figures are modified by *ja/jak*). The word order variation seems to primarily mark information structure, so it can be considered a syntactic expression of the LOC vs. LOC-EX contrast, but it also means that the resulting typology (presented in Table 5) is significantly more complex than a simple dichotomy. As for perspective structure, it appears not to be directly reflected in the morphosyntax, or if it is reflected in word order, information structure appears to be the stronger factor (cf. (35), (25)).

Table 8 demonstrates how the groups of examples identified in Table 7 align with the LOC vs. LOC-EX distinction as defined within the core vs. periphery approach, incorporating all three parameters (definiteness, information structure, perspectivisation).

The results show that all of the examples annotated as core LOC clauses have the word order F–G–V and do not contain the *ja/jak* marker modifying the figure, while the morphosyntax of core LOC-EX clauses is much more variable. Thus, in terms of the ability to account for the formal variation in figure-ground predications, the core vs. periphery approach does not seem to be a more power-

Table 8: Correspondences between formal and functional types of figure-ground predications in the Gawarbatı sample

	F–G–V	F–G–V, <i>ja/jak</i>	G–F–V	G–F–V, <i>ja/jak</i>	F–V–G
core LOC	12	0	0	0	0
core LOC-EX	5	1	31	6	2
periphery	7	0	21	1	1

ful predictor than some of the parameters used alone, in particular, information structure.

As was noted in §4.1, in addition to indefiniteness marking and word order, Gawarbatı occasionally marks the LOC and LOC-EX distinction by the lexical identity of the predicate: the verb *nic-* ‘sit’ is used exclusively in LOC clauses, and the predicative negative particle *nahn-* is used exclusively in LOC-EX clauses. These strategies were ignored in the corpus study due to their rare occurrence, but in other languages of the Hindu Kush region, the lexical identity of the predicate often plays a much more significant role. In the next section, we present an overview of the strategies for marking the LOC vs. LOC-EX contrast in the Hindu Kush languages and pay special attention to the interaction of the predicate’s lexical identity with (in)definiteness marking and word order.

5 Figure-ground predications in the Hindu Kush region

5.1 Areal survey

The following is only a preliminary survey of marking strategies in the surrounding region. Examples are mostly drawn from grammatical descriptions, text collections of varying scope, and – to a more limited extent – own field data. Emphasis has been on identifying the more prominent strategies for upholding contrasts between LOC and LOC-EX, rather than offering any in-depth systematic analysis of each language in this respect, as few if any descriptions of languages of the region are explicit about such marking strategies, possibly with the exception of Appelgren’s (2023) study on locatives and existentials in Indo-Aryan Khowar [khow1242]. Starting with the more immediate Indo-Aryan neighbours of Gawarbatı, the geographical scope will gradually widen as to include languages of all the six more prominent phylogenetic groupings that are

represented in the Greater Hindu Kush, as shown in Figure 2. No conscious attempts have been made at regularising the written representations of the languages, and we are mostly following the transcriptions applied in each of the cited works, while the glossing in a few cases has been adjusted (and somewhat simplified) to display greater uniformity.

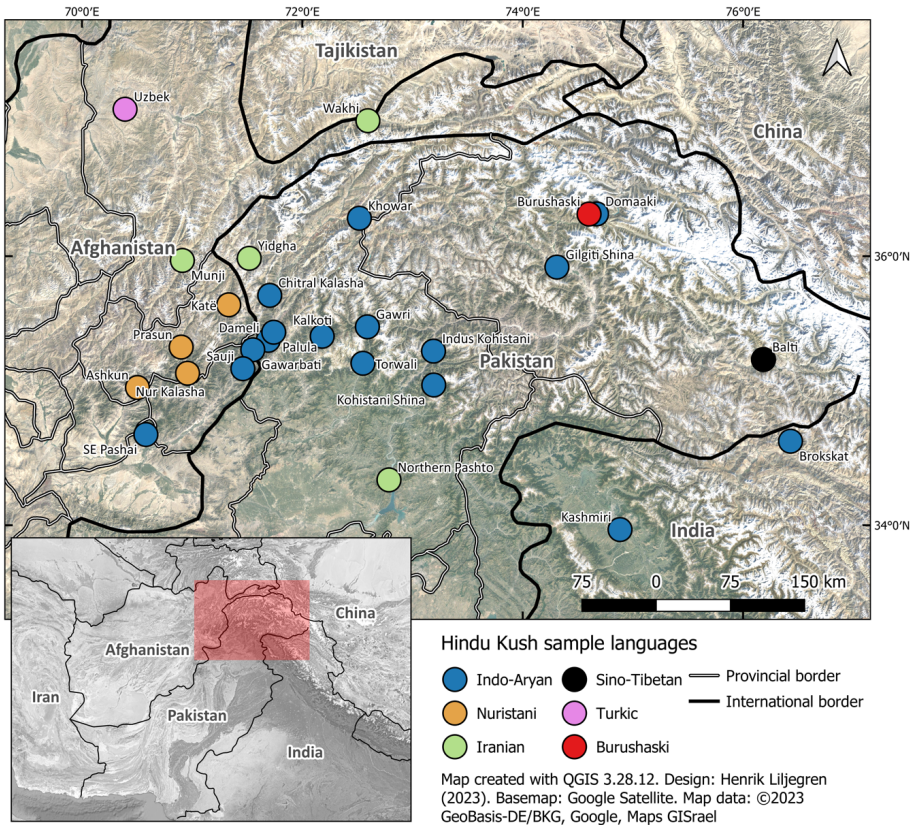


Figure 2: Languages included in the areal sample

5.2 Figure-ground predications in neighbouring speaker communities

As of today, Gawarbati speakers primarily use two other languages of wider communication in local interaction with non-Gawarbati speakers, namely Pashto (Iranian, [pash1269]) and Khovar, the former on both sides of the border, and the latter on the Pakistani side. However, historically – especially if taking the last few centuries into account – contact patterns have most likely been rather

more diverse and might not even have included those two languages, at least not to any greater extent. Among the more likely interactional candidates are — on the one hand — the upriver Indo-Aryan languages Dameli, Chitral Kalasha [kala1372] and Palula [phal1254], and downriver Sauji (closely related to Palula, [savi1242]), and possibly one or more varieties of Pashai [pash1270]. Dameli, for example, contrasts between LOC (36) and LOC-EX (37) by means of word order (F–G–V vs. G–F–V), entirely parallel to the structures in the elicited Gawar-bati material.

- (36) Dameli (Perder 2013: 131)
šey-nam bum šaa daro
 thing-PL ground on be.INAN.PRS.3
 ‘The things are on the ground.’

- (37) Dameli (Perder 2013: 124)
yee t^haan-a gram beru
 3SG.INAN place-LOC village be.INAN.PST.3
 ‘In this place was a village.’

The same word-order alternation between LOC and LOC-EX is found in comparable sentences in Chitrali Kalasha (Heegård 2006: 52, 110), Palula (HL field data), Sauji (Buddruss 1967: 66) as well as in the better described Southeastern variety of Pashai (Lehr 2014: 381, 383). In all of those languages, an utterance-final copula occurs, displaying agreement with the head of the figure NP along a variety of language-specific agreement patterns, whereas the LOC or LOC-EX identity itself does not appear to play a role in the choice of copula.

Like in Gawar-bati, an indefinite marker (derived from the numeral ‘one’) can occur in Palula (as well as in Sauji), but not obligatorily so, when the figure constitutes a part of the comment, as in (38).

- (38) Palula (B:FOY047)
haráa táaper-a ak tang yaár hín-i
 there.DIST mountain-OBL INDF narrow cave be.PRS-F
 ‘There in the mountain is a narrow cave.’

Such an indefinite marking, encoding something specific but not identifiable by the hearer, often translatable as “a certain X”, also occurs in Chitral Kalasha, as in (39), i.e. ‘in a (certain) very beautiful forest area’.

- (39) Chitral Kalasha (Heegård Petersen 2015: 114)

tasa dur ek bo çiçojak adrak-una ç-iu
 3SG.REM.OBL house INDF very beautiful forest-LOC be-PRS/FUT.3SG.NS
 ‘His house lies in a very beautiful forest area.’

Besides the ties with these neighbouring Indo-Aryan communities, Gawarbatı has been in frequent contact for a very long time with some of the Nuristani groups in the mountainous areas in the Northwest, particularly with speakers of Nuristani Kalasha [waig1243] and Katë [kati1270] (Morgenstierne 1950: 6–7, Cacopardo 2001: 241–248). Here, too, word order is the most obvious strategy for differentiating between LOC (40) and LOC-EX (41).

- (40) Nuristani Kalasha (Degener 1998: 116, 285)

toba üc uzag di [...] murāygoł-iw or-ay
 3SG.GEN well today still [...] Muraygol-LOC be-PRS.3SG
 ‘His well is even today... in Muraygol.’

- (41) Nuristani Kalasha (Degener 1998: 116, 313)

nišeygrām-iw [...] atröm-čemi bröm-čemi or-at
 Nisheygram-LOC [...] upper-villager lower-villager be-PRS.3PL
 ‘There are both up-valley and down-valley inhabitants in Nisheygram.’

Distinctly different copulas, or copular realisations, are used in Nuristani Kalasha (Degener 1998: 115–118) as well as in Katë (Halfmann 2022: 407–419), but in neither case does this appear to reflect a differentiation between LOC and LOC-EX per se. More interesting is the occurrence of posture verbs that apart from their posture usage have come to be used in copular-like ways. In Palula, the verb *bheš-* ‘sit (down)’ spans across a rather wide range of meanings. While in some cases, the posture itself is intended, in many other contexts, it should be understood as meaning nothing more specific than ‘stay’, ‘wait’, ‘settle’, or ‘be or remain present in a certain place for an extended time’, as in (42).

- (42) Palula (A:BEZ019)

eetāa ugheenīi-a bheš-í de
 there Pashtun-PL sit.down-CVB PST
 (The women had gone to pick wild herbs and brought them down to Red Plateau.) ‘Some Pashtuns were there (lit. had sat down).’

A similar meaning is expressed by Katë *žé-* ‘sit down’ in (43).

- (43) Katë (Halfmann 2022: 140)
óc kómbřom jě-něžēm
 there Kómbřom.LOC sit.down-PROG.PST.1SG.M
 ‘I was staying in Kómbřom (Kāmděš).’

To what extent this is related to a LOC vs. LOC-EX differentiation is still very much an open question. In Palula, posture verbs are primarily (if not exclusively) used in LOC-EX, and Baart (1999: 126–127) explicitly points out the use of a copula preceded by perfective adjectival participles of the verbs ‘sit’ (see (44)), ‘go’, ‘build’, and ‘adhere’ to introduce new participants in the discourse in Gawri/Kalam Kohistani [kala1373], an Indo-Aryan language spoken in the Panjkora and Swat valleys (to the East of Chitral and the Gawarbat-speaking area). He remarks that this predicate “is rather empty, and the main purpose of the clause is to assert the existence of the participant referred to” (Baart 1999: 126).

- (44) Gawri (Baart 1999: 127)
kāṇō rākā čor dākūān bāy-āl thu
 ridge on four robbers sit-ADJ.PTCP be.PRS.M
 ‘There were four robbers sitting on the ridge.’

Although ‘sit’ appears to be the more common posture verb to display this particular usage, as also seen in the Nuristani Ashkun [ashk1246] example in (45), again with an indefinite specific figure NP, an analogous use of ‘lie’ can be seen in the Nuristani Prasun [pras1239] example in (46).

- (45) Ashkun (Morgenstierne 1929: 233)
zaṅgal-wa pə kūč a pakīra niši-nestəgə
 forest-OBL in middle INDF ascetic sit-PFV.PST.3SG
 ‘There was an ascetic in the middle of the forest.’

- (46) Prasun (Buddruss & Degener 2017: 222)
üştyöb-pan zəma atinyog əžə
 tree-on snow lie.PASS.PTCP be.PRS.3SG
 ‘There is snow on the tree.’

5.3 Figure-ground predications in the wider surrounding region

When widening the scope even further, geographically and phylogenetically, the same pattern prevails as far as the correspondence between word order and the LOC vs. LOC-EX contrast is concerned, as can be seen in the Burushaski examples

in (47) and (48), LOC and LOC-EX respectively. Apart from in Burushaski,¹¹ this general pattern is evidenced in Indo-Aryan,¹² Nuristani,¹³ Iranian,¹⁴ Sino-Tibetan,¹⁵ and Turkic¹⁶ alike, as far as the wider Hindu Kush region is concerned.

- (47) Burushaski (Berger 1998: 179)
in basí-ul-o b-ái
 he garden-LOC-ESS COP-3SG.HM
 ‘He is in the garden.’

- (48) Burushaski (Yoshioka 2012: 184)
gán-ul-o han bar-čil-an b-ilú-m
 way-LOC-ESS one valley-water-INDF.SG COP-3SG.Y-NPRS
 ‘There was a stream on the way.’

The presence of an enclitic or suffixal indefinite or singulative marker *-an* (probably derived from the numeral *han* ‘one’) in (48), marking an indefinite specific figure NP in the LOC-EX, is not unique to Burushaski. It occurs in quite a few of the neighbouring Indo-Aryan varieties — at least in Gilgiti Shina ([gilg1242] *ek* ‘one’; =*ek* INDF, Radloff & Shakil 1998: 21–22), Kohistani Shina (*ek* ‘one’; =(e)*k* INDF, Schmidt & Kohistani 2008: 75–79), Indus Kohistani ([indu1241] *ék* ‘one’; =*uk* INDF, Lubberger 2022), Kashmiri ([kash1277] *akh* ‘one’; =*a:(h)* INDF, Koul & Bhat 2014: 94) and Domaaki ([doma1260]; *ek* ‘one’; =(e)*k* INDF, Weinreich 2011: 181–182) — as well as in Tibetan Balti (Sino-Tibetan, [balt1258]; *čik* ‘one’; =*či* INDF, Biemeier 1985: 86, 162), as duly noted by Yoshioka (2012: 47). Neither is the co-occurrence of the suffix and a preceding “full” numeral ‘one’ a phenomenon restricted to Burushaski but has been noted in particular by Lubberger (2022) for Indus Kohistani, and who describes at length the various uses of the indefinite specific enclitic =*uk* in this language: a) to introduce participants/entities

¹¹Burushaski (Berger 1998, Yoshioka 2012).

¹²Brokskat ([brok1247], Ramaswami 1982, Sharma 1998), Chitral Kalasha (Heegård 2006, Heegård Petersen 2015), Dameli (Perder 2013), Gawri (Baart 1999), Gilgiti Shina (Radloff & Shakil 1998), Indus Kohistani (Lubberger 2014, 2022), Kalkoti ([kalk1245], own data), Kashmiri (Koul & Wali 1997), Khowar (Appelgren 2023, Bashir 2023), Kohistani Shina ([kahi1248], Schmidt 2006, Schmidt & Kohistani 2008), Palula (own data), Southeastern Pashai (Lehr 2014), Torwali ([torw1241], Lunsford 2001).

¹³Ashkun (Morgenstierne 1929), Katë (Halfmann 2022), Nuristani Kalasha (Degener 1998), Prasun (Buddruss & Degener 2017).

¹⁴Munji ([munj1244], Morgenstierne 1938), Northern Pashto (David 2014), Wakhi ([wakh1245], Bashir 2009, SanGregory 2018), Yidgha ([yidg1240], Morgenstierne 1938).

¹⁵Balti (Biemeier 1985, Jones 2009, Read 1934).

¹⁶Uzbek ([uzbe1247], Sjoberg 1963).

(2022: 287), as exemplified in (49), b) to introduce a subset of an already known entity (2022: 291), and c) to encode approximation (2022: 293). While the enclitic =*uk* implies specificity, the preceding *ék*, also encoding indefiniteness, remains neutral in respect to specificity (2022: 297), and, when occurring alone, it often carries a generic meaning.

- (49) Indus Kohistani (Lubberger 2022: 287)
qasá gài gài ék baačàa=uk āās
 story go.PFV.F go.PFV.F INDF king=INDF be.PST.M.SG
 ‘Once upon a time there was a king.’

It is entirely plausible that this parallel use actually originates in Burushaski and has subsequently spread through language contact; alternatively it is a substratal feature in speaker communities whose ancestors shifted from Burushaski to an Indo-Aryan language. An argument in favour of the latter, rather than the former, is the existence of a plural indefinite suffix *-ik* in Burushaski (Berger 1998: 43–44) lacking paradigmatic parallels in all of these other languages except for Domaaki (Indo-Aryan, [doma1260]; Weinreich 2011: 182), a language that definitely stands at the receiving end of contact-induced change vis-à-vis Burushaski and Gilgiti Shina (Backstrom 1992: 80–83, Weinreich 2008: 314–315).

We concluded that there is relatively little evidence in the immediate neighbourhood of Gawarbatī of a correlation between a LOC vs. LOC-EX differentiation and the preference for one or the other copula or copula-like verb used in such predications. However, such preferences can be found with more certainty in the wider region, particularly in the Western part of the Hindu Kush. Iranian Pashto (now a close neighbour, but more peripheral in a historical sense) is a case in point. While an inflected copula, used in other non-verbal predications, typically occurs in LOC predications and displays agreement with the figure NP, as in (50), a separate non-inflected existential particle/copula *šta* (or *šte*) is instead used in LOC-EX predications, as in (51).

- (50) Northern Pashto (David 2014: 308)
zmā noṭ-una pə kitābče ke di
 my note-PL in notebook.OBL in be.CONT.PRS.3PL.M
 ‘My notes are in the notebook.’

- (51) Northern Pashto (Tegey & Robson 1996: 93)
pə kor ke worə šta
 in house in flour EXIST
 ‘There is flour in the house.’

In yet another set of languages, an explicit (existential) copula occurs, like in Pashto, in LOC-EX predications, but can be left out altogether in LOC predications. This is the case in Iranian Wakhi, in (52), and Munji (Morgenstierne 1938: 145), as well as in Turkic Uzbek, as in (53). In the first-mentioned language, pronominal clitics with copular functions can co-occur with an overt copula to emphasise existence in LOC-EX, while in LOC both may be (but are not always) absent (Bashir 2009: 841–842).¹⁷

- (52) Wakhi (SanGregory 2018: 126)
ja mɔl-əf ku ra ʃad
 DEM flock-PL all down.to sheep.pen.ACC
 ‘The flocks are all in the sheep pen.’

- (53) Uzbek (Sjoberg 1963: 122)
ona-m taʃkent-ta
 mother-1SG.POSS Tashkent-LOC
 ‘My mother is in Tashkent.’

This clearly contrasts with the obligatory use of copulas in Tibetan Balti (in the East). Although Balti has more than one copula verb, *yot* is the one that is normally used in LOC as well as in LOC-EX, apart from also occurring in some adjectival predications and in possessive constructions (Read 1934: 35–37, Jones 2009: 28–29). In (54), the use of *yot* in a LOC-EX is immediately followed by its use in a LOC.

- (54) Balti (Jones 2009: 71)
hundar-iŋ kʰoŋ-i kʰar yot, ŋai apičo-e
 Hundar-LOC 3PL-GEN palace EXIST 1PL.EXCL.GEN grandmother.HON-GEN
astano saŋ ina hundar-iŋ yot
 tomb also there Hundar-LOC EXIST
 ‘In Hundar there is their palace. Our grandmother’s tomb is also there in Hundar.’

All of the languages exemplified so far, regardless of their phylogenetic identity, are canonically V-final. Thus, any major word order alternation is the one observed between figure and ground only. The only obvious exception to V-finality in the wider region, as defined here, is Kashmiri, normally considered a verb second word order (V2) language, exceptional both in terms of its areal affinities and

¹⁷It should be noted that not everyone analyses the Wakhi clitics as being pronominal and/or copular, see SanGregory (2015).

in relation to other Indo-Aryan languages (Verbeke 2013: 175). However, as far as contrasting LOC and LOC-EX, that is usually obtained by an F–V–G vs. G–V–F alternation (Koul & Wali 1997: 338).

5.4 Summary

This rough areal survey, covering 27 language varieties (including Gawarbatī), belonging to six phylogenetic groups, has shown that there is an overwhelmingly frequent correlation between LOC predications and the constituent order F–G, and between LOC-EX and the order G–F. This is not to say that other word orders are absent, only that the canonical order observed in Gawarbatī has very clear areal parallels, both in the immediate geographical surroundings and in the wider Hindu Kush region, just like SOV is the only basic word order pattern represented in the region (with the marginal exception of the V2 pattern in Indo-Aryan Kashmiri). This correlation holds regardless of the phylogenetic affiliation.

More extensive research and the availability of larger language-specific corpora would be essential in order to detect and explain more exact correlations between word order and the parameters discussed in §4, but a few key observations have also been made in regard to indefiniteness marking and the predicate's lexical identity. Indefinite markers, often derived from the numeral 'one', frequently occur with the figure NP of LOC-EX predications. The semantics of such NPs is often one of indefinite specificity. A subset of languages, particularly those spoken in the eastern part of the Hindu Kush, have developed a suffixal or enclitic marker, in addition to a preposed numeral/indefinite article, possibly modelled on, or parallel to, a grammaticalisation process originating in Burushaski.

Posture verbs, most often based on 'sit', alternate with the use of more generally used copulas, particularly in the Indo-Aryan and Nuristani languages in the vicinity of Gawarbatī. Although that, too, would need more careful investigation and larger corpora for a more authoritative interpretation to be made, the initial impression is that such posture verbs occur more often in LOC-EX predications than in LOC predications, in the languages that have them, i.e. the opposite pattern to the one noted in Gawarbatī. Another observation made is a tendency in Iranian and Turkic, in the Northwest of the region, to encode LOC and LOC-EX in lexically distinct ways, such as using different copulas or completely leaving out the copula, primarily in LOC predications.

6 Conclusions

In this paper, we analysed the morphosyntactic variation in figure-ground constructions in Gawarbati and other languages of the Hindu Kush region — languages which, in terms of Creissels' (2019b) typology, lack a dedicated LOC-EX construction that is distinct from a LOC construction. In identifying the LOC vs. LOC-EX status of the clause, we mostly relied on the translations into English where LOC and LOC-EX clauses are formally distinct. In the case of Gawarbati, we also conducted a more fine-grained analysis of the corpus data, attempting to relate the observed morphosyntactic variation in the sample to the semantic/pragmatic criteria discussed in the literature as differentiating LOC and LOC-EX clauses cross-linguistically (definiteness of the figure, information structure and perspectivisation).

The investigation of Gawarbati presented in this paper is, to our knowledge, the first systematic corpus-based study of figure-ground predications in a language which lacks formal marking dedicated to differentiating LOC and LOC-EX clauses. Moreover, it is one of only a few corpus-based studies of figure-ground predications in general (another example is Budzisch 2025). In the process of annotation of natural spoken data, two remarks, relevant to the general discussion on the LOC vs. LOC-EX distinction, have been made. First, it has been shown that the majority of figure-ground predications occurring in natural discourse can be clearly assigned to the LOC type or to the LOC-EX type regardless of the criteria we choose for defining the LOC vs. LOC-EX contrast (definiteness, information structure or perspectivisation). In one third of the examples, however, the choice of the definition becomes crucial as different criteria give different results. Second, the annotation of the perspective structure in affirmative clauses turned out to be quite challenging. Thus, although defining the LOC vs. LOC-EX contrast through perspectivisation might seem promising, in practice the perspective structure in affirmative clauses, where the presence of the existence presupposition is less clear, is difficult to determine.

The corpus-based analysis, complemented by the comparison with areally close languages, allowed us to verify some of the claims made in the earlier literature and make several new observations. Overall, the results suggest that there are three morphosyntactic parameters which reflect, to varying degrees, the LOC vs. LOC-EX status of the predication in the Hindu Kush: word order, indefiniteness marking and the lexical identity of the predicate. The word order variation reflects the LOC vs. LOC-EX alternation in a very consistent way across all languages studied in this paper. The consistency of this association is likely due to the similarity of word order patterns in the Hindu Kush region and is thus a

phenomenon of areal nature. In all discussed Hindu Kush languages (except V2 Kashmiri), the unmarked word order is SOV, and in all of them, word order is flexible, allowing for the use of different word order patterns in figure-ground predications. As argued in the previous literature (Freeze 1992: 556–557), SOV languages usually have the word order F–G–V in LOC clauses and the word order G–F–V in LOC-EX clauses; the results of our study confirm these generalisations. What has been less discussed in the literature, however, is the precise mechanism behind the link between the LOC vs. LOC-EX status and word order. The corpus study of Gawarbati has shown that word order primarily encodes the information structure of the clause: for example, when the figure is topical, it appears at the beginning of the clause, and when the figure is focused, it appears in the preverbal position. Furthermore, it means that word order reflects a whole set of information-structural patterns available in figure-ground constructions which are not always easily matched with a simple LOC vs. LOC-EX opposition. Given the uniformity of word-order-related processes in the Hindu Kush, it is reasonable to suggest that, upon closer examination, a similar conclusion could be drawn for other languages discussed in this paper.

The (in)definiteness marking of the figure shows slightly more variation. In all Hindu Kush languages, we observe some kind of indefinite marking of nouns, based on the numeral ‘one’. When the figure has an indefinite marker, the clause is likely to be a LOC-EX predication, but, as shown in the corpus study of Gawarbati, this only works in one direction: far from all LOC-EX predications have the indefinite marker. In Gawarbati, the indefinite marker has many restrictions. In particular, it appears only in indefinite specific contexts. Therefore, it is a very poor predictor of the LOC vs. LOC-EX status, and, most probably, the same applies to the indefinite markers in some of the neighboring languages, e.g. Dameli and Palula. It is important to note, however, that in some languages, primarily in the Eastern Hindu Kush with a possible source in Burushaski, the numeral ‘one’-based marking may be doubled: the indefinite article preceding the noun may co-occur with the cognate suffix or enclitic attached to the noun on the right side. In Indus Kohistani (Lubberger 2022), the clitic marker has the function of introducing the participant to the discourse, a function that is often associated with LOC-EX clauses (cf. Croft’s (2022) term “presentational location” for LOC-EX). This might suggest that some languages have developed a slightly more specialised means of marking the LOC-EX clauses than general indefinite marking.

The lexical identity of the predicate is the least stable feature differentiating LOC and LOC-EX clauses in the Hindu Kush languages. First, some variation can be seen in the use of posture verbs in figure-ground predications, a phenomenon widely attested cross-linguistically (Ameka & Levinson 2007) and present in the

Indo-Aryan and Nuristani languages. While in some languages, e.g. in Palula, Gawri and Ashkun, posture verbs in figure-ground predications can serve as indicators of the LOC-EX status of the clause, in other languages, e.g. in Gawarbati, the rare copula-like uses of the same verbs are found exclusively in LOC clauses. Second, individual languages allow omission of the copula in LOC clauses and/or have specialised existential predicative markers in LOC-EX clauses. The more complex structure of LOC-EX predication, while not typologically surprising, is not a prevalent feature throughout the Hindu Kush region, but rather is associated with non-Indo-Aryan languages of the western part of the area.

To sum up, word order and general indefinite marking are more stable features in the area, but at the same time they are less specialised in terms of marking the LOC vs. LOC-EX distinction. In contrast, introductory marking of the figure and special existential verbs in LOC-EX are less stable and more idiosyncratic features in the area, but they represent a more advanced stage of grammaticalisation of the LOC vs. LOC-EX contrast. Within this view, Gawarbati appears to represent the least developed stage of grammaticalisation of the LOC vs. LOC-EX distinction, in an areal comparison.

Abbreviations

In running text

F	figure	O	object
G	ground	S	subject
LOC	locative predication	TAM	tense, aspect, mood
LOC-EX	locational-existential predication	V	verb
NP	noun phrase	V2	verb second word order

In glossed examples

The following abbreviations are not found in the Leipzig Glossing Rules:

ADD	additive	INAN	inanimate
CONT	continuous	NPRS	non-present
ESS	essive	NS	non-specific
EXIST	existential particle or copula	PTCL	particle
HM	human masculine gender	REM	remote
HON	honorific	Y	y-gender (Burushaski-specific class)

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